GOSIM Paris 2026

Topic

Accelerating the Distributed Ecosystem: AI-Powered Scaling for Oniro

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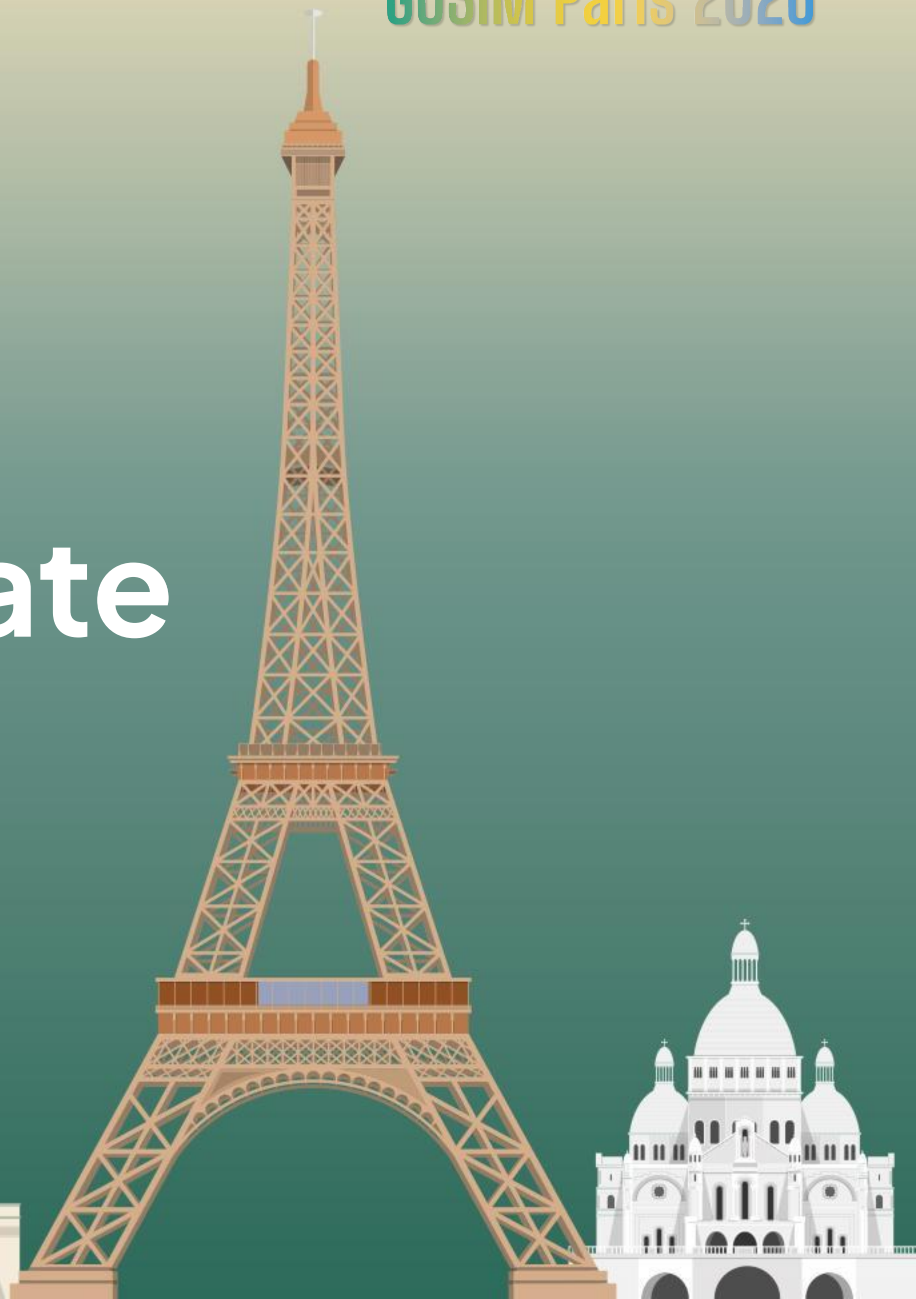
A Blue Ocean to Navigate



The Eclipse Foundation

The Eclipse Foundation

Governance & stewardship

Guiding vendor-neutral development & ensuring sustainability

IP management & licensing

Delivering IP confidence through due diligence & commercially-friendly OSI licenses

Branding, marketing & events

Building project brands, driving awareness, and hosting developer conferences and community events

Infrastructure & security

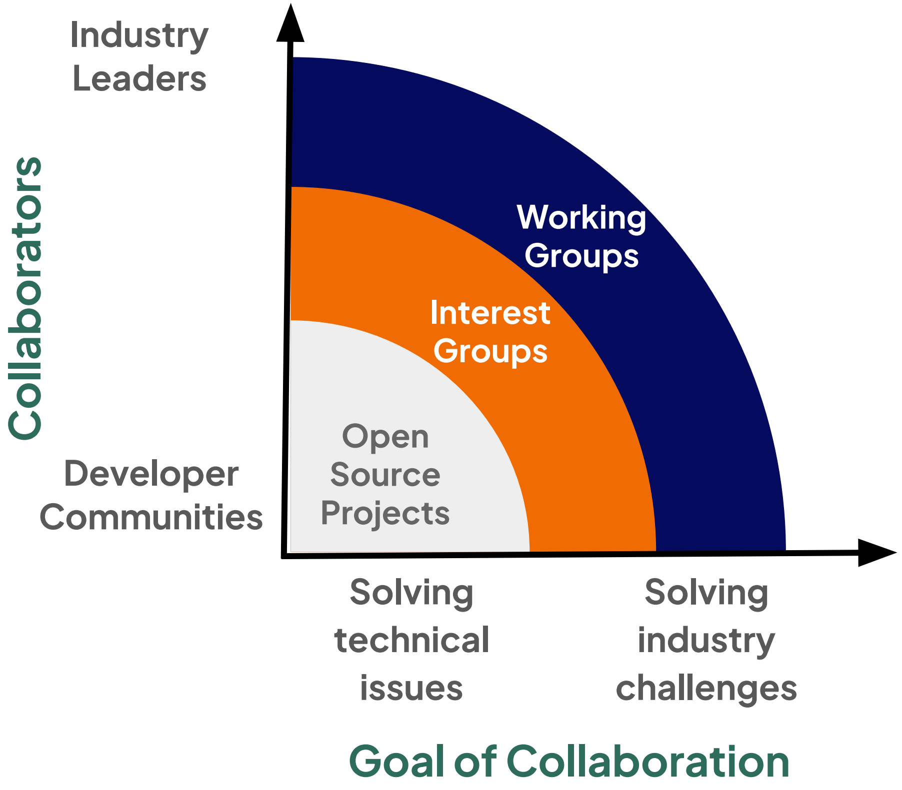
Providing reliable IT services and implementing supply chain security best practices

Community development & participation

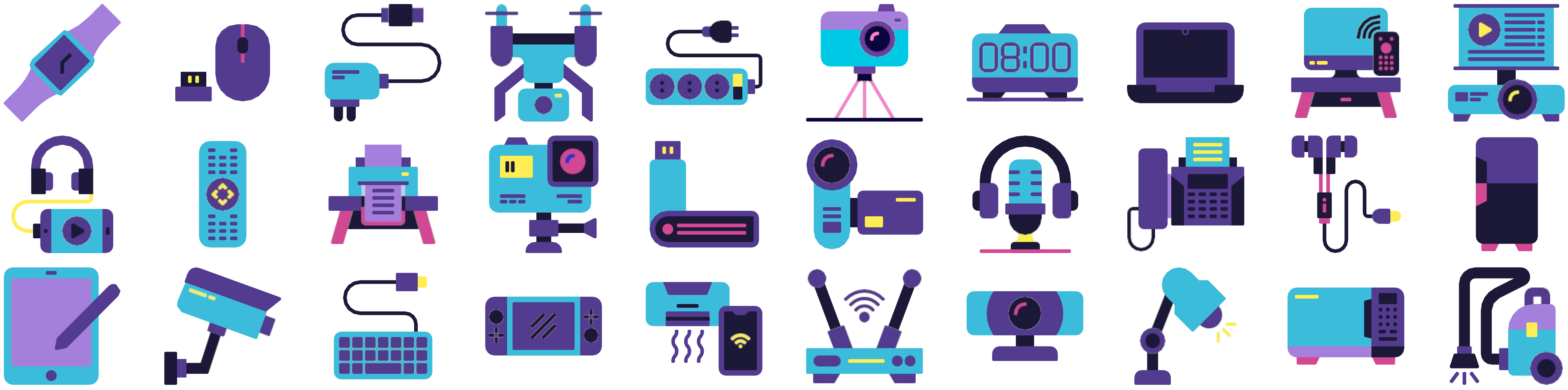
Building diverse communities & driving contributions

Fundraising & ecosystem development

Through Working Groups, raising funds and recruiting member organisations



Device fragmentation



It makes the promise of consistent software unreliable

Roots and impact

Roots

Vast hardware diversity. Millions of new, distinct models every year. Same OS, but different CPUs, GPUs, cameras, and proprietary components.

Slow updates. Once a new OS version is released, it must be modified, tested, and approved for each individual device model.

Customisation. Heavy customisation of the user interface (UI) and own proprietary services on top of the base OS.

Impact

Developers. Waste of time testing and debugging their apps against dozens of different HW and OS combinations to ensure the app doesn't crash.

Manufacturers. Prolonged time-to-market due to extensive testing requirements for each new HW component and OS update.

Users. Inconsistent user experience and lack of access to the latest security features.

The chicken-and-egg deadlock

> The 'App gap' is a math problem we're solving with velocity

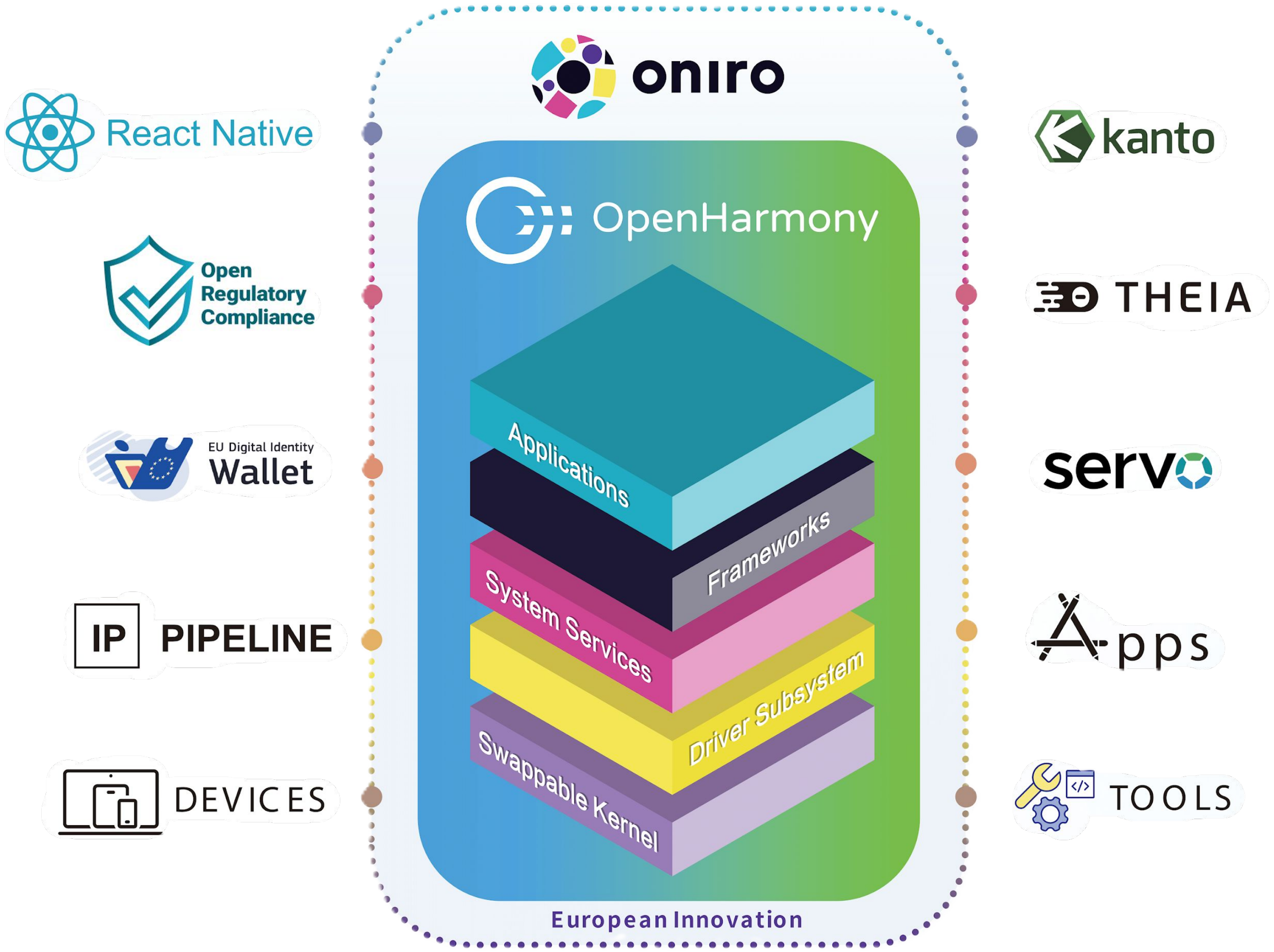
OpenHarmony is an open source framework and platform for the operating system of smart devices. It is hosted by the OpenAtom Foundation in China.

Oniro is an OpenHarmony compatible distribution focused on EU markets, and it's hosted by the Eclipse Foundation, which is based in Europe.

Oniro bridges the gap between:

- Geographical spectrum
- Different cultural paradigms
- Diverse tooling
- Broad regulatory environment
- Different processes and procedures

From kernel to applications, and beyond



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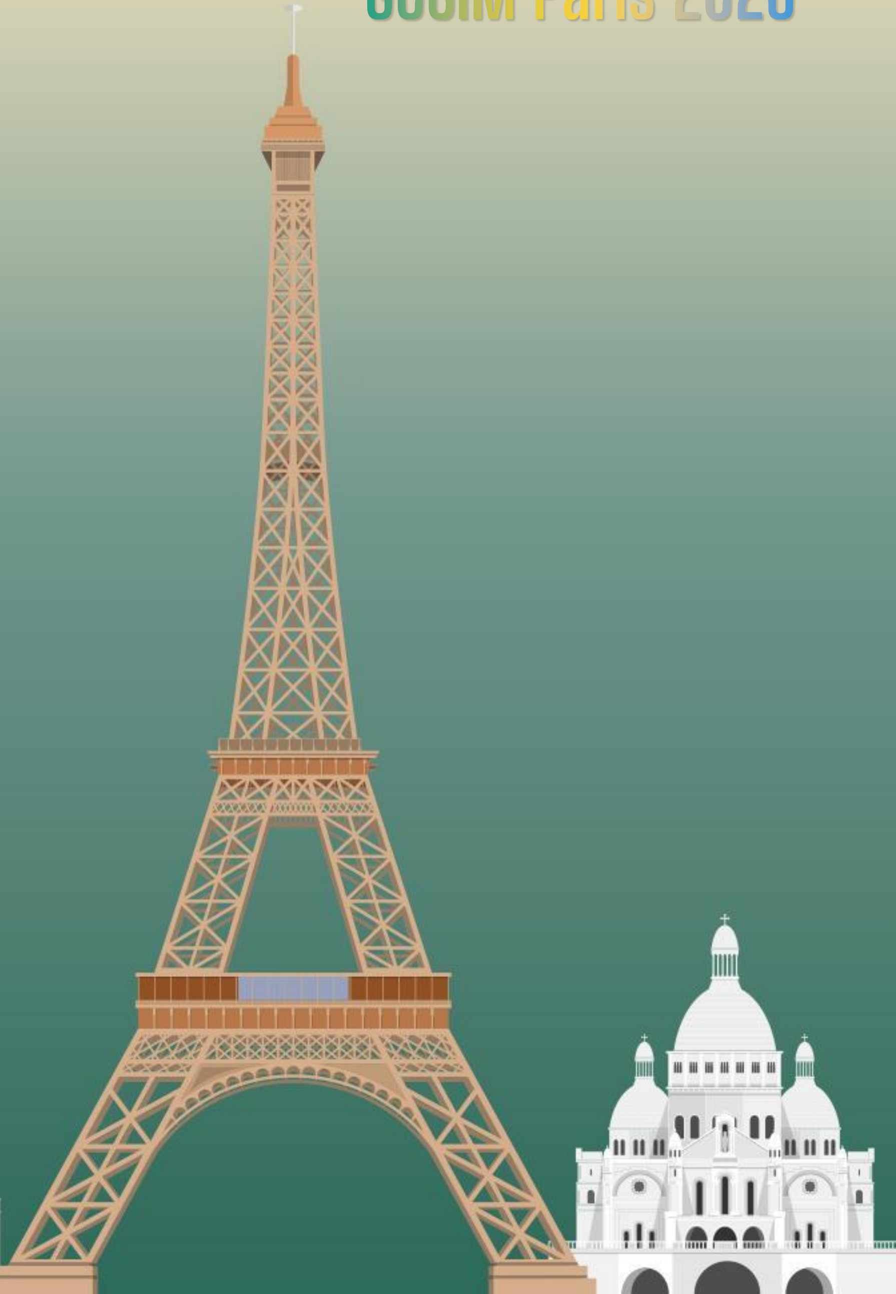
02

The Engine of Velocity: AI-Driven Engineering



Cross-platform Frameworks

+

App Ecosystem

+

Porting to Existing Devices

The future of software engineering is about orchestration, not just authorship

Lowering the entry barrier

The Necessary Shift

This is a decisive break from AOSP toward an independent stack. OpenHarmony and Oniro represent a true "Blue Ocean" for global open source OS platforms.

How do we close the app gap if entry requires a total native rewrite?

The Cold-start Problem

Millions of developers won't abandon codebases to rewrite everything in ArkTS from scratch.

Framework Liberty

Port dominant frameworks like React Native, Ionic/Capacitor and Tauri directly to the OS.

> *We don't ask developers to learn a new language; we meet them where they already work.*

Capacitor and Tauri

Beyond the Usual Names

While many look at Flutter or React Native, we focused on the two frameworks that represent the most popular philosophies in modern development.

The Web-Based Path: Ionic & Capacitor

It takes standard web technologies and puts them into a native container.

The fastest way to instantly unlock thousands of existing web-first applications for a new OS.

The Performance Path: Tauri

Uses a web interface for UI but utilises Rust for the heavy lifting.

Maximum speed and security while keeping the app size incredibly small.

Validation & Deep Dive

Maturity to handle simple web apps to high-performance software.
Catch up with Francesco Pham (Oniro PMC Lead) and learn about engineering challenges.

Developer Experience and Ecosystem Validation

Developer Experience First

Familiar Tools

We built "CLI Adapters" so developers can use the commands they already know (like `npx` or `cargo`).

Zero Learning Curve

No need to learn a new build system; the tools handle the complex "translation" to the new OS automatically.

The Lichess Validation

Not a Toy Demo

We successfully ported this heavy, production-grade application.

Real-World Complexity

It handles real-time graphics, sound, and storage with native performance, proving the architecture is ready for the "big leagues."

Speed: Weeks, Not Quarters

The AI Accelerator

These architectural ports were completed in a fraction of the traditional time.

A New Paradigm

By orchestrating AI agents to digest documentation and debug code, we've moved from theoretical research to a functional reality.

Hardware Validation: Porting Oniro to Volla devices

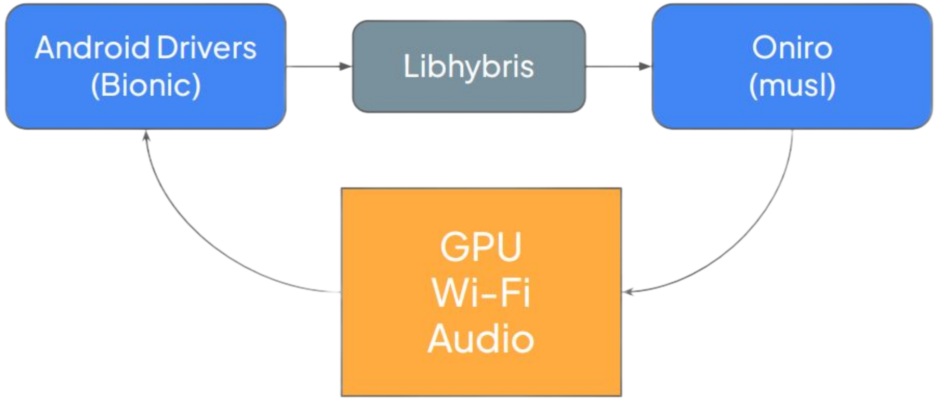
The Hardware Challenge

Most smartphones are built for Android, making their hardware drivers "speak" a language incompatible with a standard open source core.

The Bridge & AI

We utilised libhybris as a universal translator between Android drivers and Oniro.

AI agents mapped complex dependencies, moving from technical friction to a functional system.



Consumer Ready

Oniro runs smoothly on the Volla X23 smartphone and Volla Tablet with full graphical support.

This proves Oniro isn't just for development boards: it’s ready for existing commercial devices.

The multi-agent workflow

Phase 0: Research

The Foundation

Context is everything. We start by creating massive "Strategy Documents" that serve as the AI's single source of truth before a single line of code is written.

Phase 1: Decomposition

AI fails at "big ideas" but excels at "tiny tasks." We break high-level goals into small steps with clear exit criteria.

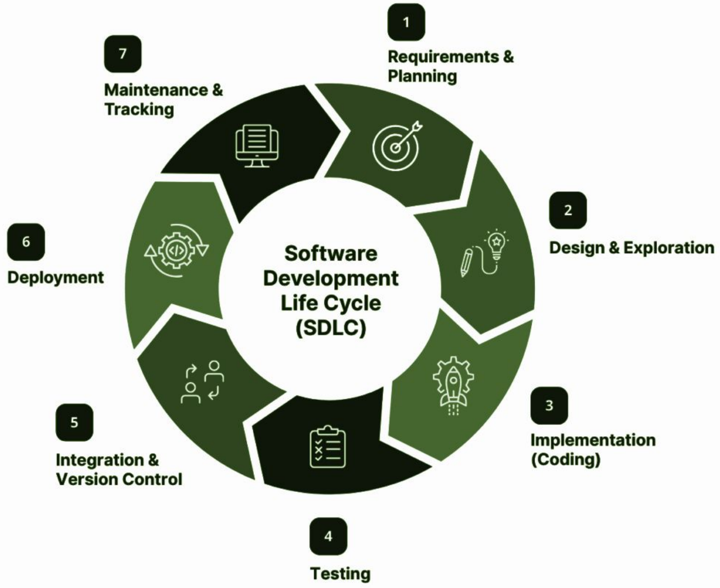
Phase 2: Self-Healing

The Loop

The AI acts as an active troubleshooter, executing a cycle: Build → Deploy → Analyse → Fix. It resolves strict-mode violations in ArkTS and mismatches autonomously.

The Human "System Intuition"

Humans solve mysteries AI can't see. When logs look "correct" but the screen is blank, intuition identifies API mismatches that save projects.



Documentation is Oxygen

Updating the docs ensures the AI learns from every mistake, making the ecosystem smarter for the next dev.

BMAD is an AI-driven framework helping you build software from ideation to agentic implementation.

The 4-Phase Guided Workflow

1. Analysis
Brainstorming and research to define the core "Brief."

2. Planning
Turning ideas into a Product Requirements Document (PRD).

3. Solutioning
Designing the architecture and blueprint before coding.

4. Implementation
Building the software piece by piece, Epic by Epic.

The Toolbox: Simple & Accessible

Node.js & Git
Industry foundations for version control and setup.

AI-Powered IDE
Using professional tools like Cursor or Claude Code.

A Simple Idea
A project vision is the only true prerequisite to start.

BMAD – Roles and Skill IDs

Analyst (Mary)

`ID: bmad-analyst`

The "Deep Researcher." Mary analyses documentation to produce the strategy (PRD) that keeps the project grounded.

Architect (Arthur)

`ID: bmad-architect`

The "System Designer." Arthur creates technical blueprints defining component interaction.

Developer (Dan)

`ID: bmad-developer`

The "Builder." Dan translates blueprints into clean, functional, and performant code.

Scrum Master (Sam)

`ID: bmad-sm`

The "Orchestrator." Sam breaks down goals into bite-sized tasks and manages priority.

UX Designer (Ursula)

`ID: bmad-ux`

The "Experience Advocate." Ursula ensures an intuitive journey and functional UI alignment.

QA (Quentin)

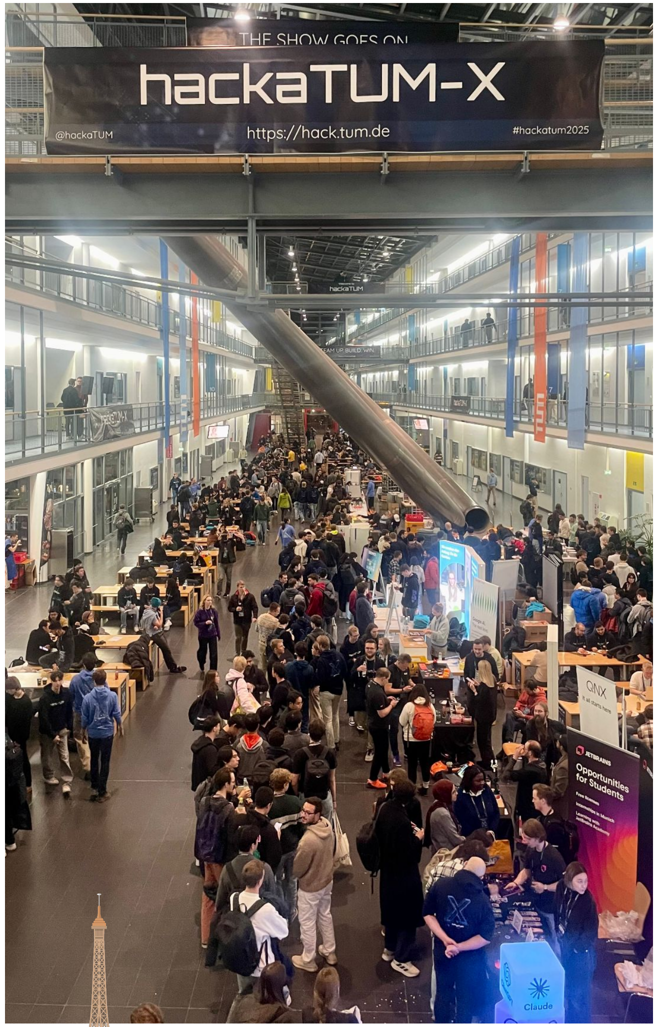
`ID: bmad-qa`

The "Gatekeeper." Quentin tests implementations to ensure requirements are met without regressions.

The Key Takeaway: Specialisation as a Guardrail

By using specific Skill IDs, we ensure the AI stays in character, creating a professional system of checks and balances where roles don't bleed into one another.

Validation - HackaTUM



03

Compliance and Integrity as our Strategic Sidecar

CRA – Regulatory compliance at its core

01

Regulatory landscape its getting more and more sophisticated.

02

The interlink among all the different regulations impacts differently depending on the use case and the industry the technology is applied to.

03

Embedding regulatory requirements into projects it's essential to facilitate market adoption.

The Eclipse Foundation as an open source SW steward

01

Monitors regulatory requirements and participates in legislative consultations to promote open source as a solution to comply.

02

Supports projects with the necessary assets to facilitate its market adoption: cybersecurity policy, vulnerability reporting, etc.

03

Cooperates with the manufacturers contributing to the technologies the Eclipse Foundartion is stewarding to develop compliance assets.

ECLIPSE™ FOUNDATION

01

The Initiative: Open Mobile Integrity Alliance

Led by Volla to establish Unified Attestation: a transparent, open source implementation of device integrity checks.

02

The Barrier: Google Play Integrity

A security service verifying if a device is "safe".

Only Google-certified OSes pass, creating vendor lock-in that blocks non-Android based OSes.

03

The "Black-Box" Trust Problem

Critical transactions rely on closed systems.

Dominant OSes include data harvesting; FOSS apps cannot guarantee privacy if the underlying OS is a "black box".

04

The Opportunity: Oniro & OpenHarmony

Ecosystem Sovereignty
Independent layer of integrity.

Market Readiness
Security confidence for high-value apps.

Privacy by Design
Transparent, safe foundation.

Closing

01

The Blue Ocean Opportunity

Moving beyond the AOSP duopoly to build an independent, sovereign software stack.

A fresh start for global open source, offering innovation and market independence.

02

AI Velocity & Solidity

The Engine

AI agents and the BMAD method reduce build time from weeks to days.

The Sidecar

CRA Compliance & Unified Attestation ensure European standards and integrity.

03

Validation & Growth

The On-Ramp

Framework Liberty for frictionless code migration.

Real-World Proof

Validated by commercial devices, porting of complex apps, and HackaTUM success.

Thank you